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## Module 3 — Decision-Making with Excel IF

### Learning outcomes

Students will be able to:

- apply Excel functions within a tabular calculation;
- write single and nested IF formulas;
- combine IF with AND and OR;
- classify student marks into A–E categories; and
- test a formula exactly at every decision boundary.

### 100-minute sequence

| Minutes | Activity                                         |
|---------|--------------------------------------------------|
| 0–10    | Inspect a grade table and identify its decisions |
| 10–25   | Review functions and calculate a weighted mark   |
| 25–40   | Single IF and comparison operators               |
| 40–55   | Nested IF and condition order                    |
| 55–75   | Build an A–E grade classifier                    |
| 75–84   | Test boundaries and invalid inputs               |
| 84–97   | Three-question short exercise                    |
| 97–100  | Checklist                                        |

**Minimum product:** a student-grade table that calculates a weighted result, pass status, and A–E grade correctly at every boundary.

### 1. Single IF

The general structure is:

`=IF(condition,value_if_true,value_if_false)`

For a pass threshold of 60:

`=IF(B2>=60,"Pass","Not yet passed")`

The `>=` is important. With only `>`, a mark of exactly 60 would be placed in the wrong category.

## 2. Combining conditions

A mark is valid only when it lies between 0 and 100:

```
=IF(AND(B2>=0,B2<=100),"Valid","Invalid")
```

A student requires remedial work when the mark is below 60 **or** attendance is below 75%:

```
=IF(OR(B2<60,C2<75%),"Remedial","No remedial")
```

Use AND when every condition must be true. Use OR when any one condition is sufficient.

## 3. Nested IF for A–E grades

Use these practice rules:

| Range  | Grade |
|--------|-------|
| 80–100 | A     |
| 70–<80 | B     |
| 60–<70 | C     |
| 50–<60 | D     |
| 0–<50  | E     |

Validate the overall domain before classifying:

```
=IF(OR(B2<0,B2>100),"Invalid",  
  IF(B2>=80,"A",  
    IF(B2>=70,"B",  
      IF(B2>=60,"C",  
        IF(B2>=50,"D","E")))))
```

Where supported, IFS can improve readability:

```
=IF(OR(B2<0,B2>100),"Invalid",  
  IFS(B2>=80,"A",B2>=70,"B",B2>=60,"C",B2>=50,"D",TRUE,"E"))
```

Conditions are checked from the highest boundary downward. Once a true condition is found, later conditions are not used.

## 4. Practice — student-grade table

Create these columns:

| Column | Content                         |
|--------|---------------------------------|
| A      | Student name                    |
| B–D    | Assignment, midterm, final exam |
| E      | Final mark                      |
| F      | Validity status                 |
| G      | Letter grade                    |

Use the practice weighting 30% assignment + 30% midterm + 40% final exam:

```
=B2*30%+C2*30%+D2*40%
```

Then classify the result. Test at least these final marks:

-1, 0, 49, 50, 59, 60, 69, 70, 79, 80, 100, 101

These categories are learning data, not an official academic grading policy.

## 5. Common errors

- excluding an exact boundary, for example using  $>80$ ;
- starting conditions at the lowest threshold, so every higher mark enters the first category;
- storing numbers as text;
- assigning a letter grade to a value outside 0–100; and
- using a long nested formula without documenting its rules.

For rules that change often, a lookup table is usually easier to maintain than deeply nested IF statements. Treat that method as an extension.

## 6. Boundary-test table

Before trusting the formula, write expected and actual outcomes:

| Input | Expected | Actual  | Pass?   |
|-------|----------|---------|---------|
| 49    | E        | fill in | fill in |
| 50    | D        | fill in | fill in |
| 59    | D        | fill in | fill in |
| 60    | C        | fill in | fill in |
| 79    | B        | fill in | fill in |
| 80    | A        | fill in | fill in |
| 101   | Invalid  | fill in | fill in |

Testing one ordinary value does not prove that all boundaries are correct.

## Short exercise — 3 questions

### 1. Predict

What is the result of this formula for  $B2=80$ , and why?

`=IF(B2>80,"A",IF(B2>70,"B","C"))`

### 2. Fix

Repair the formula so A begins at  $\geq 80$ , B at  $\geq 70$ , and any input outside 0–100 returns Invalid.

### 3. Explain

A student receives 100 for assignments, 50 for the midterm, and 50 for the final exam. Predict the final mark and grade. Explain why testing only a mark of 75 is insufficient.

1. The result is B because  $80 > 80$  is false and  $80 > 70$  is true.
2. `=IF(OR(B2<0,B2>100),"Invalid",IF(B2>=80,"A",IF(B2>=70,"B","C")))`.
3. Final mark:  $30+15+20=65$ , therefore C. A test at 75 covers only a middle branch; every exact boundary, domain endpoint, and invalid value also needs a test.

## Checklist

- ☐ I can write single and nested IF formulas.
- ☐ I deliberately choose > or >=.
- ☐ I validate the domain before classifying.
- ☐ I test exactly at every boundary.

## Further reading

1. Michael Alexander & Dick Kusleika, *Microsoft Excel 365 Bible*, 2nd ed., Wiley, 2025.
2. Wayne L. Winston, *Microsoft Excel Data Analysis and Business Modeling*, 6th ed., Microsoft Press, 2019.
3. Bernard Liengme & Keith Hekman, *Liengme's Guide to Excel 2016 for Scientists and Engineers*, Academic Press, 2019.
4. E. Joseph Billo, *Excel for Scientists and Engineers: Numerical Methods*, Wiley, 2007.

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